

EDMS 27-301-1
SPECIFICATION
FOR
Vented Nickel Cadmium Batteries

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يتم بدء العمل بالمواصفة الحالية بتاريخ ٢٠٢٢/٥/١

** توجد بنود إختيارية (Option items) يجب تحديدها بواسطة شركة التوزيع قبل الطرح.

EEHC DISTRIBUTION MATERIALS SPECIFICATION

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EEHC DISTRIBUTION MATERIALS SPECIFICATION**EDMS 27- 301 - 1****30/7/2021****1. SCOPE:**

This specification covers the minimum technical requirements for design, engineering, manufacturing, and testing of vented nickel cadmium batteries used in medium voltage primary substations and similar applications. The system requires a reliable and uninterrupted D.C. supply for supplying D.C. Power to emergency lights, closing and tripping coils of circuit breakers, relays ...etc.

2 - APPLICABLE STANDARDS:

Unless otherwise specified in this specification, the vented nickel cadmium batteries should comply with the latest edition of IEC standard and should be designed, manufactured and tested in accordance with the applicable IEC standards or approved equivalent including but not limited to the following:

IEC 60623	Secondary Cells and Batteries Containing Alkaline or other Non-acid Electrolytes - Vented Nickel-Cadmium Prismatic Rechargeable Single Cells
EN 50272-2	Safety requirements for secondary battery installation
IEC 62485-2	Safety requirements for secondary batteries and battery installations – Part 2: Stationary batteries
IEC 61438	Possible Safety and Health Hazards in the Use of Alkaline Secondary Cells and Batteries – Guide to Equipment Manufacturers and Users.
IEC 60050-482	Primary & International Electro technical Vocabulary, Chapter 482 Secondary Cells and Batteries.
IEC 60417	Graphical Symbols for Use on equipment.
IEEE 1106	Recommended Practice for Installation Maintenance, Testing, and Replacement of Nickel Cadmium Batteries for Stationary Applications
IEEE 1115	Recommended Practice for Sizing Nickel-cadmium Batteries for Stationary Applications
IEEE Std 1184	IEEE Guide for the Selection and Sizing of Batteries for Uninterruptible Power Systems

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The performance of vented nickel cadmium batteries should be guaranteed for the following environmental conditions, any differences in the guaranteed performance should be clearly set out in the offer.

Minimum ambient temperature	-5°C
Maximum ambient temperature	45°C (50 °c as option).
Maximum relative humidity	95%.
Maximum altitude	1000 m

4- CONSTRUCTION REQUIREMENT:**4.1 Electrodes**

The positive and the negative electrodes shall be of pocket plate type where the active materials shall be retained in pockets formed from perforated Nickel-plated steel strips.

4.2 Electrolyte

The electrolyte shall be an aqueous solution of potassium hydroxide (KOH) and lithium hydroxide (Li OH). For permanent temperatures below -20°C a special high-density electrolyte shall be used. There is no replacement of electrolyte during the battery service life.

4.3 Separator

Separate the plates and insulate the plate frames from each other and they allow free circulation of electrolyte between the plates.

4.4 Container

The cell containers should be either:

- For standard type, a translucent polypropylene, The electrolyte level is then visible through the side walls. Min. and Max. electrolyte level marks shall be displayed on the two sides of the container. Thermal welding should be used between container parts without additional substances (such as glue).
- The cell lid shall be welded to the cell container, gluing is not allowed. The cell container shall be single cell container design for individual replacement of 1.2 V cells. Block types with higher voltages are not allowed.

4.5 Venting System

Each cell shall be equipped with flame-arresting top vents for easy maintenance of cells. The vent cap should allow topping up of water.

Recombination feature by adding external gas recombination vents is not acceptable for safety reasons.

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4.6 Terminals & Connections

- The intercell connectors should be made of nickel-plated copper or insulated tin-coated solid copper bar.
- Positive and negative terminals must be marked by color washers as follows:
 - Red for the positive pole, (+) marking.
 - Blue for the negative pole, (-) marking.
- In order to protect the battery terminals from external accidental short-circuits resulting from any metallic item falling on top of the battery, components such as inter cell connections, insulating covers inter-row connecting cables, and collecting bars shall be IP21 compliant (tested and approved according to EN 60529 Ed2.1 Degree of protection provided by enclosure (IP code).
- To avoid accidental short Circuit All Connecting bar between Cells shall be insulated with suitable insulating materials, and Cell connecting bolts shall be with insulated head.



- Inter-row connecting cables Shall be 35 mm² cross section.
- Can be rapidly recharged at a high rate for next duty.

5- BATTERY CHARACTERISITICS:

- No of cells/battery 92.
- Nominal voltage 1.2 v/cell.
- Max. voltage/cell 1.4 v.
- Nominal voltage 110 V/set.
- Max. voltage 128.8 V/set.
- Rated capacity $C_5 = 100$ A.H. (120 A.H. **acc. to ...EDC request**), supply variable load, its type is (KPL) vented nickel cadmium(K), plastic container (P) and low rate of discharge (L).
- The battery bank shall be able to provide upon discharge a current of 0.2C A for 5 hours to an end voltage of 1 V/cell.
- The cell can tolerate complete discharge with almost no permanent deterioration of capacity or life.
- Max short circuit current $10 \cdot C_5$ Amps.

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- Charging characteristics: The battery can be charged by all following methods: -
 - 1-**Constant current charging:**
 - a) Standard charge 0.2 C5 for 7-8 hours.
 - b) Fast recharge 0.3 C5 for 2.5 hours followed by: 0.2 C5 for 2.5 hours.
 - c) Trickle charge at about 1-2 m. A/A.H. capacity.
 - 2-**Constant voltage:**
 - a) Float voltage range at about 1.38-1.47 V/cell.
 - b) Boost voltage range at about 1.47-1.73 V/cell.
- The batteries shall be complete with the mounting racks.
- **Itemized separate price should be submitted for the following accessory items: -**
 - 1.(1) Conventional type Hydrometer to show density value.
 - 2.(1) piece digital voltmeter.
 - 3.Thermometer from 0°C - 100°C.
 - 4.Two di-Electric Grease Cans.
 - 5.A (20) meters length of a 35 mm² flexible copper cable used to connect the battery to the charger, its insulation should be withstanding the heat and chemical liquid leaking to which the batteries are exposed.
 - 6.A (4) pieces of 35 mm² copper lugs.
 - 7.The purity of copper lugs should be not less than 99.9 %.
 - 8.A (2) group of numbering labels; these labels should be water proof.
 - 9.A (15 meter) 16 mm² flexible stranded copper cable used for earthing the racks (yellow and green color) and a (2) pieces of copper Lug 16 mm².
 - 10. The battery racks should be made of (an anti-rust and corrosion) material to protect the racks from both alkaline materials leaking and operating conditions, as well as the racks must be strong enough to bear the weight of the batteries.
 - 11. Also; the racks should be provided with the following: -
 - Supporting legs (to raise the racks above the ground level by 10 cm), these supporting legs should be insulated with a suitable material to withstand the chemical liquid leaking and water.
 - Two ground bolts (one in each side of racks).
 - 12. The rack should be consisting of 4 steps for carrying the whole battery bank and isolated from ground by supported isolators.
 - 13. In case of two racks delivered for one battery set. The supplier should provide a 35 mm² flexible copper cable with necessary number of lugs: to maintain the connection between the inter- row connection in the 2 racks. (**acc. to ...EDC request**)

6- GENERAL:

- **Accessories:** Bolts, washers, leads, necessary for connecting cells in series, should be delivered.

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- **Storage:** Batteries should be delivered empty and can be stored for a long period under specified conditions without affecting final application requirements.
- **Operation:** Battery completely charged will act as stand-by D.C. supply in a floating condition, parallel to a battery charger and the distributor supplied with D.C. current, Also the battery should be able to withstand full charging and discharging currents at an electrolyte temp. up to 55 ° C.
- The batteries shall have an intended minimum design life of at least 20 years.
- The technical offer should include the no of times of charging & discharging for battery age.
- The batteries shall be designed for operation on continuous-float basis.
- The batteries shall be with a stabilized self-discharge rate not exceeding 3 percent per month at +20 °C after the initial period having relatively low rate of discharge.
- The batteries shall be designed to prevent cracking, warping or bulging of containers/covers due to plate growth or differential expansion of plate supports, during normal service life.
- The batteries shall be with shock absorbing, heat resistant, plastic material for container and cover. The container shall be transparent and have minimum and maximum level markings for electrolyte.
- The batteries shall be with gastight and electrolyte tight sealed joints to provide a spill-proof and leak-proof construction including for cell terminal.
- The impedance of the cell is nearly constant from a fully charged state to an almost completely discharged state
- The batteries shall be transported "empty and discharged".
- Capacity loss not exceeding 1.5% per year.
- The specific gravity of the alkaline electrolyte ranges from 1.160 to 1.250.
- Battery can be charge to 200 % without damage. (Should be tested during FAT test for 10 sec.)

7- MARKING:

- a. For each battery (on both side) shall be permanently and legibly marked in English with the following information:
 - a. Battery type.
 - b. Cell Nominal Voltage.
 - c. Cell Capacity.
 - d. electro- lied density.
- b. For modular steel tray/rack shall be permanently and legibly marked in English and Arabic with the following information:
 - a. Manufacturer's name/country of origin.
 - b. Date of manufacture (month and year).
 - c. Date of commissioning at site.
 - d. Manufacturer's type or model number.
 - e. Applicable standard.
 - f. Ampere-hour capacity and the number of hours on which the rating is based corresponding to 20 °C and minimum end-of-discharge cell voltage of 1V.
 - g. Number of cells.

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h. Individual cell voltage (Nominal).

i. Recommended float voltage and boost/equalizing voltage per cell at 20 °C.

j. Purchase Order No./Contract No./Job Order No.

8- TESTS:

All tests shall be carried as per the relevant standard, tests shall comprise of type tests and acceptance tests.

8.1 Type Tests:

The bidder shall furnish the following type tests reports along with the offer.

- Verification of constructional requirement.
- Verification of marking.
- Verification of dimensions.
- Test for capacity.
- Charging Test as per IEC 60623.
- Discharging test as per IEC 60623.
- Test for voltage during discharge.
- Ampere Hour and Watt – hour efficiency test.
- Test for loss of capacity on storage.
- Endurance test.

8.2 Acceptance Test:

These tests shall be performed in presence of ...EDC representative from the sample selected from the lot offered for supply.

- Verification of constructional requirement.
- Verification of marking.
- Verification of dimensions.
- Test for capacity.
- Test for voltage during discharge.
- Ampere Hour and Watt – Hour efficiency test.

9- DRAWING and CATALOGUE:**9.1 Offer Documents:**

The following documents should be submitted with the offer: -

- All necessary drawings including layout, dimensions.
- Technical catalogues containing the technical data.
- Installation, operation and maintenance instructions.

9.2 Site Documents:

The Supplier should submit the following documents soft & hard copy with the batteries to the site- :

- All necessary drawings including layout, dimensions.
- Technical catalogues containing the technical data "maximum internal resistance value
- for battery Cell, maximum resistance for connecting bar between Batteries, charging Curve and methods for charging the Batteries with constant voltage method."

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- Installation, operation and maintenance instructions.

- guarantee schedule for nickel cadmium batteries.
- Manufacture Acceptance Test Reports.

10- GUARANTEE:

- The supplier should guarantee the nickel cadmium batteries against all defects arising out of faulty designed or workmanship or of defective materials for a period of 3 years from putting in service.
- The good quality and manufacturing of the materials should be guaranteed.
- Any parts which on account of poor quality of the materials, manufacturing defects or poor assembly, should be replaced or repaired in the shortest time possible and free of charge.
- If the battery becomes defective within the initial interval of service life, it will usually be replaced by the battery manufacturer at no charge.

11- GUARANTEE TABLES:

- The tenderer should fill in the attached guarantee tables.
- Any tender not accompanied with clear and complete guarantee tables should be rejected.

12- POTASSIUM HYDROXIDE (KOH) SPECIFICATION:

- KOH		88.0 %
- K ₂ CO ₃	not more than	1.5 %
- CL	not more than	0.02 %
- SO ₄	traces	
- PO ₄		Nil
- N	not more than	0.01 %
- CA		Nil
- FE ₂ O ₃ + AL ₂ O ₃		0.03 %
- FE		0.005 %
- HEAVY METALS		Nil
- NAOH		1.5 %
-Net weight of one box		12.5, 25 kg.

ملحوظة :

- يجب أن يكون سعر البطاريات شاملة المياه الخاصة بالبطاريات حيث سيتم توريد البطاريات جافة ويتم توريد المياه منفصلة عنها .

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GUARANTEE SCHEDULE For NICKEL CADMIUM BATTERIES

ITEM	Supplier
1. Manufacturer's name	
2. Standard specifications applied	
3. Model / Type	
4. No. of cells forming the battery bank	Cells
5. Nominal voltage / cell	(V dc/cell)
6. Battery Bank Nominal Voltage or D.C. System Rated Voltage	(Vdc)
7. Minimum Permitted System Voltage	(Vdc)
8. Maximum Permitted System Voltage (during equalizing)	(Vdc)
9. Recommended float charging voltage at 20 °C	(V dc/cell)
10. Equalizing Voltage	(V dc/cell)
11. End of Discharge voltage	(V dc/cell)
12. Open circuit voltage in charged condition	(V dc/cell)
13. Guaranteed service life (minimum) under operating conditions	Years
14. Battery Loading a. Continuous loading b. Short duration load c. Momentary load d. Any other load	-----Amp. for ----hours -----Amp. for ----hours -----Amp. for ----hours -----Amp. for ----hours
15. Battery Amp. hours Capacity as computed based on 5 hours discharge rate (including all correction factors)	AH
16. Protection degree	
17. Cross section of inter- cell connector	mm ²
18. Battery terminal short circuit current	A
19. Specific gravity @ 20 °C	
20. Time for total discharge during short circuit	Sec.

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21. Maximum permitted ripple voltage content for float and equalizing charge	
22. Maximum discharge rate a. Discharge duration b. Discharge current	minutes/seconds A
23. Max. recommended maximum recharge current following discharge	A
24. Float Charging Current at the recommended float voltage setting	mA/AH
25. Maximum self-discharge rate per month at 20 °C	
26. Maximum capacity loss	% Per year
29. maximum allowable deep discharge/ percentage of discharge depth	
30. Material of container	
31. Weight of the cell	Kg
32. Dimensions of cell / size according to the drawing (H, W and L) in mm	mm

We guarantee the data given above for the equipment offered.

Signature:

Date: